

AgriShield – SIH Feature Deliverables Report

AI-Powered Plant Doctor, Outbreak Intelligence & Agronomic Advisory Platform

SIH MILESTONE

Date: September 13, 2026 Platform: Full-Stack Web + Local AI Engine + PWA Status: All Features Verified & Tested (100% Pass)

1. 🗨️ Local Ollama Qwen2.5 3B AI Integration

Offline-Capable Generative Agronomic Advisor (/api/chat)

AI ACTIVE

Direct connection to local Ollama inference server running **Qwen2.5 3B** on CPU with zero internet reliance.

Key Engineering Highlights:

- Native HTTP client in backend/ollama_client.py with temperature: 0.3 & num_predict: 85 tokens.
- Inference response latency optimized to **~3.3 seconds** on local CPU.
- Zero-downtime automated fallback to curated knowledge base if Ollama is paused.

Farmer UX & Voice Features:

- Clear AI source attribution badge: Qwen2.5 3B Local AI vs Verified Knowledge Base.
- One-tap quick question chips for Rice Brown Spot, Tomato Blight, Neem Oil dosage, etc.
- Full multilingual Voice-to-Text and Text-to-Speech in 4 Indian languages.

2. 🏛️ Government Agricultural Schemes & Subsidies Explorer

Dedicated Farmer Financial Assistance & Subsidies Portal (#schemes)

CENTRAL & STATE

Direct Benefit Transfers (DBT), crop insurance subsidies, equipment grants, and subsidized credit explorer.

Scheme Name	Category	Subsidy / Benefit	Official Portal
PM-KISAN	Direct Cash	₹6,000 / year via Aadhaar-linked DBT in 3 equal installments	pmkisan.gov.in
PMFBY	Crop Insurance	1.5%–2% farmer premium; covers flood, drought & post-harvest losses	pmfby.gov.in
Kisan Credit Card (KCC)	Credit & Loans	Up to ₹3 Lakhs at effective 4% interest with prompt repayment incentive	myscheme.gov.in
SMAM Mechanization	Machinery & Drone	40% to 50% capital subsidy on tractors, power tillers, sprayers & drones	agrimachinery.nic.in
Soil Health Card	Soil Health	100% Free 12-parameter soil testing report every 2 years with NPK advice	soilhealth.dac.gov.in
PMKSY Micro-Irrigation	Water & Drip	45% to 55% subsidy on precision drip & sprinkler micro-irrigation systems	pmksy.gov.in
PM-KUSUM	Solar Pumps	Up to 60% Govt subsidy (Center + State) for standalone solar water pumps	pmkusum.mnre.gov.in

Includes category filter pills, instant search, required documents tags, and direct toll-free helpline dialers.

3. 📅 Multi-Day Crop Health Recovery Timeline

14-Day Chronological Plot Rehabilitation Protocol (#plants & AI Detect)

CROP LIFECYCLE

Actionable, day-by-day agronomic trajectory guiding farmers from initial disease diagnosis to complete vegetative recovery.

Day 1: Sanitation

Isolate diseased foliage; burn/bury infected plant debris. Switch off overhead sprinklers to prevent spore spread.

Day 3: Spray Action

Apply targeted organic/copper spray (Mancozeb 2.5g/L or Neem Oil 1500ppm) between 06:30–08:30 AM calm window.

Day 7: Lesion Arrest

Inspect halo margins. Active yellow margins turn to dormant brown rings. Apply foliar potassium booster.

Day 14: Clearance

Emergence of lesion-free new apical shoots. Retake AgriShield AI leaf scan to confirm healthy status.

Supported for fungal blight, bacterial blight, yellowing nutrient chlorosis, and healthy prophylactic maintenance.

4. 🚨 Outbreak Alerts & Community Disease Intelligence Map

🚨 Nearby Outbreak Alerts

PROXIMITY ENGINE

- Haversine radius calculation (/api/outbreaks/nearby).
- Automatic farmer dashboard warning banners for active clusters within 15 km.
- Containment protocols and certified spray advisories.

🗺️ Community Disease Map

GIS VISUALIZER

- Interactive Leaflet.js map with OpenStreetMap tiles (/api/outbreaks/map).
- Severity-coded clusters (Red: Critical, Orange: High, Yellow: Moderate).
- Quarantine buffer rings displaying containment radii in km.

5. 🌤️ Weather-Based Disease Risk Predictor

Micro-Climate Risk Forecasting (/api/risk/weather-forecast)

MICRO-METEOROLOGY

- 0–100 Dynamic Risk Index:** Computed from hourly temperature, relative humidity, and precipitation.
- Night Dew Condensation Window:** Detects hours when relative humidity exceeds 85%, creating spore germination films.
- Optimal Spray Window:** Flags calm morning periods (<12 km/h wind, zero rain probability) to prevent chemical wash-off.
- 5-Day Risk Trajectory:** Forecasts upcoming risk trends with High/Moderate/Low badges.

6. 🌐 4-Language Localization & Full-Stack System Architecture

🌐 Multilingual Engine

4 LANGUAGES

Every feature, scheme, advice text, button, and navigation item is fully localized in **English, Hindi (हिन्दी), Tamil (தமிழ்), and Marathi (मराठी)** in frontend/i18n.js.

🏗️ Production Stack

FULL-STACK

FastAPI backend + dual-write persistence (SQLite for relational records + MongoDB 8.3 for telemetry) + PyTorch CNN + local Ollama LLM + PWA offline cache.